Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

THIRD SEMESTER B.TECH. (Data Science)

END SEMESTER EXAMINATIONS, December 2024

MATHEMATICAL FOUNDATION FOR DATA SCIENCE [MAT-2139]

REVISED CREDIT SYSTEM

Time: 9:30-12:30pm **Date: 02/12/2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

Q.NO	Questions	Marks	CO	PO	BTL
1A.	The Median and Mode of the following wage distribution are known to be Rs. 33.50 and Rs. 34 respectively. Find the values of f_3, f_4 and f_5 . Wages (in Rs): 0-10 | 10-20 | 20-30 | 30-40 | 40-50 | 50-60 | 60-70 | Total Frequency: 4 | 16 | f_3 | f_4 | f_5 | 6 | 4 | 230	**3**	***1***	***1,2,8, 12***	***1,2,3,4***
1B.	Find the measure of skewness from the following table giving the wages of 230 people. Wages (In Rupees) | Number of People 70-80 | 12 80-90 | 18 90-100 | 35 100-110 | 42 110-120 | 50 120-130 | 45 130-140 | 20 140-150 | 8	***3***	***1***	***1,2,8, 12***	***1,2,3,4***
1C.	Calculate the first four moments of the following distribution about the mean and hence find β_1 and β_2. x: 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 y: 1 | 8 | 28 | 56 | 70 | 56 | 28 | 8 | 1	***4***	***1***	***1,2,8, 12***	***1,2,3,4***

2A.	Fit a second-degree polynomial to the following data by the method of least square method for the given data set X: 0, 1, 2, 3, 4 Y: 1, 1.8, 1.3, 2.5, 6.3	***3***	***1***	***1,2,8,12***	***1,2,3,4***
2B.	In a certain town 40% have brown hair, 25% have brown eyes, 15% have both brown hair and brown eyes. A person is selected at random. (i) If he has brown hair, then what is the probability that he has brown eyes also. (ii) Determine the probability that he neither have brown hair nor brown eyes.	***3***	***2***	***1,2,8, 12***	***1,2,3,4***
2C.	$If\ X$ and Y are two-dimensional random variables having joint pdf $f(x,y) = \begin{cases} \frac{1}{8}(6-x-y), & 0<x<2\ ,2<y<4 \\ 0, & \text{otherwise} \end{cases}$ Then i) find $P(X<1\|Y<3)$ ii) $P(X+Y<3)$	***4***	***3***	***1,2,8, 12***	***1,2,3,4***
3A.	A two-dimensional random variable (X,Y) is uniformly distributed in the region bounded by the circle $x^2+y^2=a^2$, Find Marginal probability distribution of X and Y.	***3***	***2***	***1,2,8, 12***	***1,2,3,4***
3B.	The number of accidents in a year to taxi drivers in a city follows a Poisson distribution with mean 3 out of 1000 taxi drivers. Find approximately the number of the drivers with i) no accident in a year ii) more than 3 accidents in a year.	***3***	***2***	***1,2,8, 12***	***1,2,3,4***
3C.	The average marks in an entrance examination is 76 and the standard deviation is 11. a) The institution provides a special scholarship for all the students who score marks above 90. What is the percentage of students who are eligible for the scholarship? b) It is known that 40% of the students have failed in the exam, what is the cut off for failure? c) Calculate the percentage of failure if the cutoff for failure is 70?	***4***	***3***	***1,2,8, 12***	***1,2,3,4***
4A.	Let $\bar{X}$ and s^2 be the mean and variance of a random sample of size 25 from the distribution N (3,100). Evaluate $Pr\{0<\bar{X}<6\ ,55.2<s^2<145.6\}$.	***3***	***3***	***1,2,8, 12***	***1,2,3,4***
4B.	Let X be a random variable whose probability density function is given by	***3***			

	$f(x) = \begin{cases} e^{-2x} + \frac{1}{2}e^{-x} ; x > 0 \\ 0 \quad ; otherwise \end{cases}$. 1) Write down the moment generating function for X. 2) Using this moment generating function to compute the first and second moment.		***4***	***1,2,8, 12***	***1,2,3,4***
4C.	Suppose that (X, Y) is uniformly distributed over triangular region $R = \{(x, y); 0 \leq x < y < 1\}$. Find the marginal probability distribution function of X and Y and hence obtain the correlation coefficient between X and Y.	***4***	***4***	***1,2,8, 12***	***1,2,3,4***
5A.	Seven students were given intensive training and 5 tests were conducted in a month. The scores of tests 1 and 5 are as following Test 1: 52, 43, 52, 27, 36, 43, 61 Test 5: 63, 41, 62, 36, 31, 53, 70 Do the scores from test 1 to 5 shows an improvement? Test the significance difference at 5% Level of Significance with 1.943.	***3***	***5***	***1,2,8, 12***	***1,2,3,4***
5B.	The demand for a particular spare part in a factory was found to vary from day to day. In a sample study, the following information was obtained **Days** | **Mon** | **Tue** | **Wed** | **Thurs** | **Fri** | **Sat** **No. of parts** | 1124 | 1125 | 1110 | 1120 | 1126 | 1115 Test the hypothesis that the number of parts does not depend on the day of the week at 5% Level of Significance comparing with 11.07?	***3***	***5***	***1,2,8, 12***	***1,2,3,4***
5C.	The annual packages of students from different IIMs offering different specializations are given by the following data i) Does the annual package significantly differ among the IIMs? ii) Does the annual package significantly differ across the specializations? IIMs Specializations | **A** | **B** | **C** | **D** **Marketing** | 9.4 | 8.4 | 8.5 | 10.9 **Finance** | 10.6 | 8.8 | 11.3 | 9.8 **System** | 8.6 | 10.5 | 9.9 | 10 **Operations** | 11.2 | 10.6 | 11 | 9.3 Test at 5% Level of Significance with 3.86.	***4***	***5***	***1,2,8, 12***	***1,2,3,4***